

Vector Spaces

A *vector space* V is a non-empty set together with a binary function $+$ and unary functions c for every $c \in \mathbb{R}$ satisfying the following axioms:

- 1 $+: V \times V \rightarrow V$, that is, if $u, v \in V$ then $u + v \in V$.
- 2 $+$ is associative. $x + (y + z) = (x + y) + z$ for all $x, y, z \in V$
- 3 $+$ is commutative. $x + y = y + x$ for $x, y \in V$.
- 4 there is $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
- 5 for every $v \in V$, there is $-v \in V$ such that $v + (-v) = 0$.
- 6 for every $c \in \mathbb{R}$ and $v \in V$, $cv \in V$.
- 7 for every $c \in \mathbb{R}$ and $u, v \in V$, $c(u + v) = cu + cv$.
- 8 for every $c, d \in \mathbb{R}$ and every $u \in V$, $(c + d)u = cu + du$.
- 9 for every $c, d \in \mathbb{R}$ and every $u \in V$, $c(dv) = (cd)v$.
- 10 for every $v \in V$, $1v = v$.

abelian group
Distributivity

Ex: $\mathbb{R}^n$ is a vector space for all n

where $\underbrace{\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}} + \underbrace{\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}} = \underbrace{\begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}}$ and

$\underbrace{c \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}} = \underbrace{\begin{pmatrix} cx_1 \\ \vdots \\ cx_n \end{pmatrix}}$ for all $c \in \mathbb{R}$.

Both + and scalar mult. are closed on $\mathbb{R}^n$.

What is 0 in $\mathbb{R}^n$?

$$0 = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \checkmark \quad \text{Axiom 4.}$$

Axiom: Given $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, what is $-\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$?

$$\begin{pmatrix} -x_1 \\ \vdots \\ -x_n \end{pmatrix} \text{ acts as } - \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

$$\text{Check: } \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} -x_1 \\ \vdots \\ -x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = \mathbf{0}.$$

Axiom 5.

Check Axiom 9 for $\mathbb{R}^n$: $c, d \in \mathbb{R}$ and

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n. \quad \text{Need to show}$$

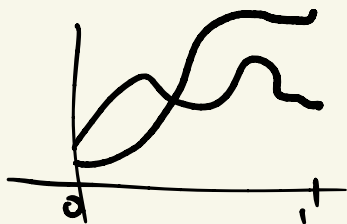
$$c(dv) = (cd)v. \quad (\text{Multiplication is associative}).$$

$$\begin{aligned} c\left(d\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}\right) &= c\begin{pmatrix} dv_1 \\ \vdots \\ dv_n \end{pmatrix} = \begin{pmatrix} c(dv_1) \\ \vdots \\ c(dv_n) \end{pmatrix} = \begin{pmatrix} (cd)v_1 \\ \vdots \\ (cd)v_n \end{pmatrix} \\ &= (cd)\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}. \end{aligned}$$

Exer: Check all the other axioms.

$\mathbb{R}^n$ with the given $+$ and scalar mult is
a vector space.

Ex: V is the set of continuous functions
on $[0,1]$.



Define $+$:

$f, g \in V$ then

$$(f+g)(x) = f(x) + g(x) \quad \text{for } x \in [0,1]$$

(Calculus - the sum of each is cont.)

If $c \in \mathbb{R}$, $f \in V$ then we define

$$(cf)(x) = cf(x) \quad \text{for } x \in [0,1]$$

Why is $+$ on this V associative and commutative?

$f, g \in V$, why is $f+g$ = $g+f$?

Evaluate both sides at every $x \in [0,1]$.

$$\begin{aligned} (f+g)(x) &= f(x) + g(x) = g(x) + f(x) \\ &= (g+f)(x) \end{aligned} \quad \checkmark$$

Axiom 4: What is 0 here?

0 is the zero function which is $0(x) = 0$ for all $x \in [a, b]$. 0 is continuous on $[a, b]$ and so is in V .

If $f \in V$ then $(f + \overset{\substack{\downarrow \text{zero function.}}}{0})(x) = f(x) + 0(x) = f(x)$

So $f + 0 = f$ and the zero function is 0 in V .

Axiom 5: For $f \in V$, what is $-f$?

$$(-f)(x) = -f(x) \quad \text{for all } x.$$

Check: $f + (-f) = 0 \leftarrow \text{zero function.}$

Small facts

Fact

In any vector space V ,

- 1 0 is unique,
- 2 for any $v \in V$, $-v$ is unique (and is equal to $(-1)v$), and
- 3 for any $c \in \mathbb{R}$, $c0 = 0$.

① If 0 and $0'$ both acted like 0 in V

then
$$0 = \underline{0} + 0' = \underline{0'} + \underline{0} = 0'$$

Subspaces

If V is a vector space and W is a non-empty subset of V then we call W a subspace of V if W is closed under $+$ and scalar multiplication i.e. if $u, v \in W$ then $u + v \in W$ and if $c \in \mathbb{R}$ and $v \in W$ then $cv \in W$.

Examples of subspaces

Given an $m \times n$ matrix A , the set of solutions of $Ax = 0$ is a subspace of $\mathbb{R}^n$ called the *nullspace* of A , written $Nul(A)$.

The span as a subspace

Suppose that $v_1, \dots, v_k$ are vectors in $\mathbb{R}^n$. The span of $v_1, \dots, v_k$ is a subspace of $\mathbb{R}^n$. Recall that the span of $v_1, \dots, v_k$ is the set of all linear combinations of $v_1, \dots, v_k$.